An Explainable AI Model for IoMT Network Attack Detection

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Abstract

The Internet of Medical Things (IoMT) has become increasingly more integrated into healthcare delivery in recent years. Along with it's many benefits in improving healthcare efficiency, it has also opened the medical industry to a wide variety of new cyberattacks, raising many concerns involving the privacy of patients data as well as their safety when receiving medical treatment. A common solution to this has been Machine Learning (ML) based Intrusion Detection Systems (IDS), which have provided high accuracy in classifying malicious network traffic in many IoT applications, and some IoMT scenarios. However, more complex ML based IDS are fundamentally less interpretable, which is an issue as it prevents trust from both patients and medical professionals. This project will address this critical gap through the design, implementation, and evaluation of a deep learning model incorporating Explainable Artificial Intelligence (XAI) that detects network attacks specific to an IoMT network.

1 Introduction

Smart and connected medical devices are increasingly integrated into the delivery of healthcare services, leading to the development of the Internet of Medical Things (IoMT), a subset of the Internet of Things (IoT) encompassing medical equipment and devices such as heart rate and cardiovascular monitoring, remote medicine alert systems, and temperature tracking. IoMT technology offers many advantages by enabling continuous monitoring of patients' health. This assists healthcare professionals in making remote diagnoses and better informing their clinical decisions; for example, data collected from IoMT devices has been used to predict patients' risk of heart disease [1]. With the growing digitalisation of healthcare systems, there has been a significant increase in cyberattacks targeting medical infrastructure. According to the HIPAA Journals' report, there have been 546 healthcare breaches that affect more than 500 people so far in 2025 [2]. This, coupled with the sensitive nature of medical data, raises major concerns about patients' privacy, where patients' personal data may be exposed, as well as potential concerns about patients' safety, given that IoMT devices are used to deliver treatments. Recent research has shown that Machine Learning (ML) can be effectively used to detect network attacks by analysing network traffic [3, 4, 5]. Compared to traditional network attack detection methods, such as signature-based systems, ML techniques offer significant improvements in detecting unknown attack types, this is because they are able to recognise complex patterns in network traffic attributed to malicious activity [6]. However, one problem faced by complex ML techniques is their lack of interpretability in decision-making; in the context of healthcare systems, this interpretability is crucial, as medical staff and patients need to trust that IoMT networks are secure. To address the lack of interpretable ML models, this project will develop a Deep Learning IDS and use (XAI) techniques to make its decisions more interpretable in IoMT systems. This ensures that users of the system can understand and trust the model while still achieving high detection accuracy. The proposed model will be trained and evaluated using the CIC2024 IoMT dataset, a publicly available dataset synthesised by the Canadian Institute for Cybersecurity [7]. This dataset has been specifically designed to develop and benchmark network attack detection models in IoMT environments, incorporating a variety of common IoMT devices and a wide variety of both benign and malicious network traffic.

2 Literature Review

In this literature, there will first be an analysis of the architecture of IoMT systems and the associated security concerns they face. Next, there will be an analysis of the existing research on Machine Learning based Intrusion Detection Systems, with a focus on how different algorithms have been implemented and tested in IoT and some IoMT scenarios. Thirdly, it discusses the role of XAI in building trust in IDS and further justifying the need for these techniques in IoMT environments. Finally, there will be a comparison and critical assessment of various available datasets for benchmarking IDS systems for IoMT.

2.1 IoMT Architecture and Security Concerns

IoT, and subsequently IoMT systems, are typically divided into three primary layers: perception, network, and applications [8]. In the perception (physical) layer, we have the actual medical sensors and wearable devices that collect information from patients, such as tracking cardiovascular and heart health, temperature, and sleep patterns. This layer faces potential vulnerabilities, with the potential for tampering and false data collection, which could ultimately lead to incorrect diagnoses based on incorrect data. The network layer enables these devices to communicate with other devices in the IoMT network using protocols such as Wi-Fi and Bluetooth. This layer is susceptible to attacks such as DoS and DDoS, which can lead to disruptions in treatment and pose an immediate risk to patients' safety when IoMT devices are unable to operate. Man-in-the-middle attacks at this layer can also lead to data breaches, compromising the confidentiality of sensitive patient and medical staff data.

The IDS discussed later in this literature review, as well as the IDS that will be developed in this project, are primarily based on this network layer. Moving to the application layer, we find software vulnerabilities where cyber threats, such as phishing and ransomware, can lead to interruptions in medical care and further data breaches.

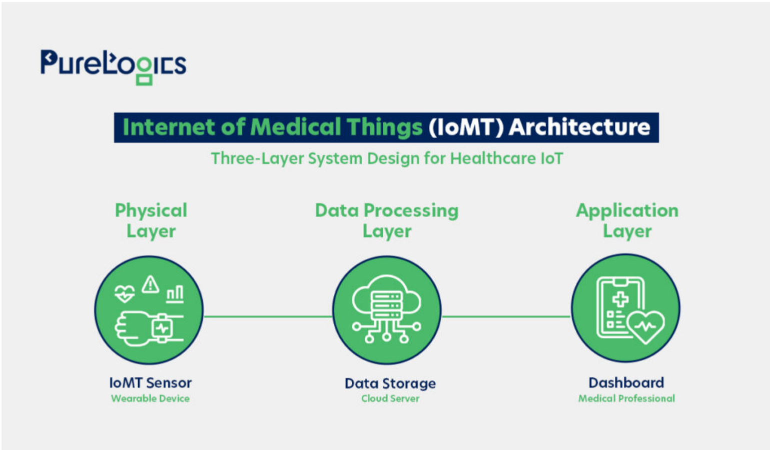

Figure 1: Illustration of layers in an IoMT system. Source: [9]

Across these layers, there are three clear points of risk for IoMT technology: patients' data privacy, patients' safety, and disruption to medical systems. The variety of areas that are vulnerable in these systems highlights the need for a more rigorous and trustworthy cybersecurity system to reduce the number of successful attacks. Effective Intrusion Detection Systems (IDS) are an essential tool for preventing the wide variety of attacks that IoMT networks face. The majority of these attacks that are detected by IDS are found between the perception and data processing layer, where data such as readings are transferred over protocols like MQTT, Bluetooth and Wi-Fi.

2.2 ML-Driven IDS

Traditional Intrusion Detection Systems (IDS), such as signature-based methods, are not well-suited to the field of IoT networks due to various unknown attack types. This has lead to research in recent years primarily being focused on hybrid Machine Learning (ML) methods for detecting network attacks, including Random Forest and SVM [10]. These ML-based systems aim to analyse large amounts of network traffic and distinguish between benign and malicious activity. This section will focus on reviewing the most well-documented models used and how their performance is evaluated in various IDS. Continuing with the shift toward ML-driven IDS approaches, early ML-based IDS explored the efficacy of algorithms such as Decision Trees and SVMs, both of which achieve promising results when tested on benchmark datasets. For example, one IDS uses a Hybrid Decision Tree tested and evaluated on the NSL-KDD dataset [11] achieves 83% accuracy and 0.83 F1 score, outperforming other ML methods such as SVM and KNN [12]. However, this model does not distinguish between attack types such as DoS, MITM, and others, which is desirable for IoMT security; the authors suggest future work to address this limitation. Deep learning methods provide a solution to this; a deep reinforcement learning model was developed and tested alongside traditional ML methods to detect malicious network traffic in IoT networks. Using the NSL-KDD dataset [11], the best performing model was the Double Deep Q-Network (DDQN), achieving an accuracy of around 80% and an F1 score of 0.91[13]., Comparing this with the Hybrid Decision Tree, it is clear that this form of deep reinforcement learning excels in attack detection; the slightly lower accuracy is acceptable as the DDQN model has potential for multi-class detection. A noticeable drawback of deep learning here is the time taken to train the model compared to traditional ML, as well as the lack of interpretability it provides. In [14], an ML IDS geared towards IoMT environments is presented. Using the ToN_IoT Network dataset [15], the authors carried out comparisons between eleven different attack classification models. The most effective classifier was found to be the Adaptive Boosting ensemble technique, achieving an accuracy of 99.18%. However, one potential gap in the implementation of this model is the dataset used; it has not been specifically created to reflect IoMT networks, but rather more general IoT networks. This could cause the model to face challenges in a real world IoMT environment, where IoMT devices make use of specific lightweight protocols such as Bluetooth and MQTT.

2.3 Explainable Artificial Intelligence (XAI) in IDS

Explainable Artificial Intelligence (XAI) refers to a set of tools that aim to make the predictions made by a deep learning model understandable to humans, this is in the form of showing the features the model decides played the biggest role in making the final prediction or classification of a model. Obviously, in a healthcare setting, healthcare professionals and patients must be able to trust that the IoMT network is being kept secure by an IDS. This requires that the IDS prevents malicious network traffic from entering the network, whilst ensuring genuine medical communications and benign traffic are not intercepted; false alarms could lead to disruptions to patients' treatment. However, we still want to accurately prevent malicious attacks that could cause reduced patient safety and sensitive medical data being compromised. Therefore, integrating XAI into a deep learning IDS allows the healthcare industry to understand and validate the classification made by the system. The two main types of XAI models are model-specific and model-agnostic. Model-specific XAI is suited to a specific group of models, meaning they offer a deep insight into the model's decision-making; however they cannot be applied to a wide variety of model types. Model agnostic techniques are the opposite, as they can be applied to many model types with less in-depth insight. [16]. Two popular examples of XAI techniques are LIME and SHAP. LIME (Local Interpretable Mode-agnostic Explanations) is a model-agnostic XAI algorithm proposed in 2016, it explains the prediction of a classifier though approximating it with a simpler and interpretable model [17]. Later, SHAP (Shapley Additive exPlanations) was introduced in 2017 [18], similar to LIME, it is a model-agnostic algorithm that determines how each feature contributes to a models output. SHAP can be computationally costly when the model it is applied to is very complex and has a large number of features. The explanations provided by LIME can differ even when the instances are similar, this is due to the random nature of it sampling, leading to less reliable results. Recent studies and research demonstrate that XAI models, such as LIME and SHAP, are effective in explaining predictions made by IDS systems. In [19], researchers successfully introduced an explainable machine learning framework for IDS. This framework was built around the SHAP method and used the NSL-KDD dataset [11] to evaluate the model's performance. However, this paper mentions the framework's limitations and highlights the need for more datasets to be tested to improve the explainability. A deep learning based IDS is proposed in [20], in which the authors integrate XAI methods, including both LIME and SHAP. This paper evaluates the performance on both the NSL-KDD dataset as well as the UNSW-NB15 datasets [11, 21], demonstrating that integrating XAI techniques into these models makes them more transparent and trustworthy. There is limited work in applying XAI techniques specifically to IoMT scenarios; addressing this gap will allow the development of more trustworthy IDS models in healthcare settings that use IoMT networks.

2.4 Evaluation Datasets and Benchmarks

The accuracy and reliability of Machine Learning (ML) based Intrusion Detection Systems (IDS) are primarily dependent on the quality and the realism of the data used to train, validate, and test the model. This is especially important in IoMT settings because the data transmitted in these networks are fundamentally more sensitive than a typical IoT network, making the need for secure communications essential. This section will critically evaluate the available datasets that have previously been used in ML-driven IDS models, discuss the limitations, and justify the selection of the CIC2024 IoMT Dataset for this projects specific focus. The dataset, NSL-KDD [11], frequently mentioned in the previous sections of this literature review, has been used extensively in training IDS models. This dataset was synthesised to improve upon its predecessor (KDD'99) [22] and has been able to produce promising results in many research applications. However, this dataset has drawbacks such as its high redundancy of its records, which can lead to poor generalization in ML models [23]. This dataset would not be suitable for the model proposed in this report because it does not reflect attacks faced in an IoMT system, and does not feature more modern attack types. Newer datasets provide significant improvements of NSL-KDD, one notable example is the CICIDS2017 dataset [24], this dataset includes more up-to-date and realistic network traffic found in IoT networks. However, this would also not be suited to IoMT systems due to it not containing common examples of network traffic, such as lightweight protocols MQTT and Bluetooth, which are used by IoMT devices. To maximise the realism of the data used to train the proposed model, this project will use the CIC2024 IoMT Dataset [25]. This dataset was specifically designed for developing and benchmarking network security systems for IoMT networks, containing 18 different attack types targetted at 40 IoMT devices [7]. It includes traffic types missing from the previously mentioned datasets including common protocols used by IoMT such as MQTT and Bluetooth. This dataset provides a wide variety of up-to-date and relevant attack vectors faced by real-world IoMT devices. The devices in the dataset can be seen in Figure 2

Figure 2: CICIoMT2024 topology. Source: [7]

The structure of the dataset will guide the development of this model. Due to the large class imbalance where a large portion of the network traffic is of type DDoS, the evaluation of the model will need to incorporate various metrics such as F1-score, Recall, and Precision to ensure the model minimises incorrect classifications.

3 Project Specification

3.1 Motivation

The use of IoMT systems continues to spread in the field of modern healthcare, with many routine health checks now able to be performed remotely and continuously. Unfortunately, with this increase of its use, there has also been an increase in the amount of cyber-attacks focused on healthcare services [2]. The main motivation of this project is to address the lack of trust that can be placed in Machine Learning based Intrusion Detection Systems, through developing an explainable model who's decisions can be verified and understood.

3.2 Aims and Objectives

The overarching goal of this project is to design, implement, and evaluate a reliable IDS designed for IoMT environments, using deep learning methods. Once this has been implemented and evaluated, XAI techniques (LIME and SHAP) will be applied to provide insight into the models' classifications of attack types. These results will then be validated and discussed to ensure the highlighted features presented are logical to each each type of attack.
The primary objectives of this project are:

- Obtain and prepare the CIC2024 IoMT Dataset, ensuring it is ready for model training, including necessary preprocessing steps to address features of the dataset such as the class imbalance.
- Design and implement the Deep Neural Network for the IDS model, then train it using the now prepared dataset, using the validation set to tune hyperparameters. Ensuring that it achieves satisfying evaluation metrics, such as F1 score, before integrating the explainable AI into the model.
- Integrate XAI techniques into the model (LIME and SHAP), then conduct analysis on the output of both methods to determine if the methods consistently flag the same features as the most crucial for predicting a given attack type. The robustness of the explainable elements can also be validated against the known characteristics of certain attacks.

Task	**Deliverable**	**Milestone**
Prepare Dataset	Preprocessed and Balanced CIC2024IoMT Dataset: The dataset is cleaned, class imbalance has been addressed, features have been encoded, split into training, validation and testing sets.	Dataset readiness: Features are correctly encoded and classes are balanced, ready for model training.
DNN IDS Model Implementation	Tuned Deep Neural Network: Model Architecture has been fully implemented and hyperparameters have been tuned.	IDS Performance Goal reached: The IDS achieves an F1 score of at least (or as close as possible) 0.8 after tuning hyperparameters.
Integrate XAI Techniques to Interpret DNN	SHAP and LIME implemented to work on the DNN's classification of various data points.	Successfully obtained several feature explanations using both LIME and SHAP, confirming the the XAI methods are working.
Validate Interpretability of XAI techniques	Analysis of the output from SHAP and LIME, validating whether they identify correct features consistent with certain attacks. Discuss any differences in explanations between both techniques.	XAI techniques successfully identify majority of features associated with attacks.

Table 1: Project Objectives

3.3 Gantt Chart Timeline

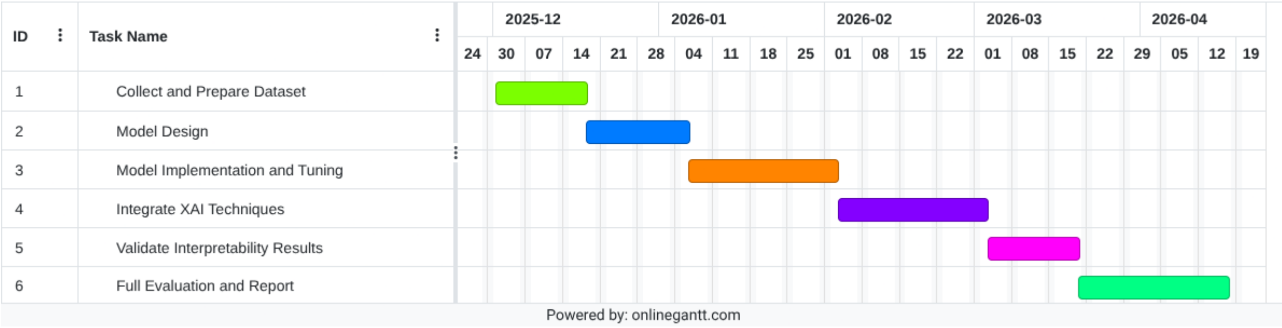

Figure 3: Gantt Chart of the project timeline

This chart shows a rough estimate for the time frame in which various stages of the project will be completed. This includes extra slippage time to tune the model as well as perfect the report and documentation of the results.

3.4 Methodology and Implementation Plan

3.4.1 Dataset

- The project will use the CIC2024 IoMT dataset, which is due to its focus on IoMT networks as well as its up-to-date nature, and the variety of IoMT relevant attack types it simulates.
- Dataset cleaning will involve the removal of any redundant features in the dataset, and create some categorical encodings for each of the different attack types.
- Address class imbalance, possibly using SMOTE to balance the dataset, as the majority class by far in the dataset is DDoS and DoS attacks.
- Split the dataset into subsets: training, validation, and testing. Ready for the process of training and evaluating the model.

3.4.2 Model Design

A Deep Neural Network is the model type that will be implemented in this IDS. This is because it was among the strongest performers when tested on the chosen dataset [7], achieving solid F1 scores in 2, 6, and 19 class classifications, narrowly being beaten by the RandomForest model. The XAI part of the project will address the lack of interpretability, which limits deep learning models in IDS. The validation set, reserved during preprocessing, will be used for hyperparameter tuning, to maximise performance measured primarily by F1 score, among other metrics like accuracy, precision and recall.

3.4.3 XAI Integration

Both SHAP and LIME will be implemented and tested on the models' classifications of network attacks. Both methods will be used to show which features were the most important when making a prediction, and will be validated by checking if these features are logical to the attack classification they are explaining, such as DoS attacks.

3.5 Evaluation and Success Criteria

IDS Model performance will be measured using F1-score, aiming to achieve a score greater than 0.85 is a reasonable target, which would indicate that the model strikes a strong balance between both precision and recall when detecting malicious network traffic. The benchmark for the proposed model will be based on existing results published in the paper for the CIC2024 IoMT Dataset [7], primarily the DNN mentioned and the Random Forest model. To measure the success of the integration of XAI techniques (LIME and SHAP), there will be a comparison between the explanations provided by both methods to see if they both identify the same features across attacks.

3.6 Risks and Ethical Considerations

The dataset that has been selected for this project does have a large class imbalance, with there being substantially more entries of DDoS and DoS network traffic compared to other attack types and benign traffic. This could lead to the model incorrectly classifying malicious network traffic as DDoS and DoS attacks, resulting in inaccurate attack detection, putting patient safety at risk if medical systems are interrupted. To combat this, the model could employ preprocessing techniques such as SMOTE (Synthetic Minority Over-sampling Technique) during training. Sourcing of the data needed to train this model is also an ethical concern, due to the data transferred on IoMT networks potentially being sensitive and private to patients. The project's use of the CIC2024 IoMT Dataset is an intentional answer to this concern, since it is a publicly available dataset that has been synthesised, and there is no handling or storage of real patients' data.

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